Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/31/2008
PI Name:	Gaboyard, Sophie Ph.D.		
Project Title:	Adaptation of Rodent Vestibular Hair Cell Neurotransmission in Altered Gravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60612-7308	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI-RFP-04-01 Postdoctoral Fellowships
Start Date:	11/01/2004	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lysakowski, Anna (MENTOR/University of Illinois, Chicago)		
Grant/Contract No.:	NCC 9-58-PF00506		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP. Space motion sickness is the earliest impairment experienced by humans in altered gravity. It is an important problem, since it severely alters performance of affected astronauts. We propose to study the early mechanisms that can affect the adaptation of mammalian vestibular hair cells in altered gravity. All specific aims will focus on utricular hair cell neurotransmission in mice. The first aim will provide an overview of synaptic transmission by looking at the vesicle recycling rates in utricle submitted to hypergravity over time. The second correlated aim will attempt to understand the time-scale of molecular mechanisms that can sustain the modification of hair cell neurotransmission in hypergravity. Both aims will provide a time-scale of the early modifications that can occur in primary gravity receptors undergoing altered stimulation. The last aim of this project is to study the functional capabilities of adult utricular hair cells whose development occurred under conditions of sensory deprivation. This last ground-based experiment will use a mammalian		

Task Description:	"weightlessness" model, the tilted mouse. This last aim will provide some insight about the risks of developing organisms in space. These objectives are directly relevant to different goals of the NSBRI Neurovestibular Adaptation team, since they can lead to the development of countermeasures to limit the risk of: 1) "disorientation and inability to perform landing, egress, or other physical tasks, especially during/after g-level changes", and 2) "possible chronic impairments of orientation or balance function due to microgravity". Centrifugation will be used to submit mice to hypergravity. Their utricular maculae will be studied using immunofluorescent staining, imaging, deconvolution and 3D reconstruction. A precise map of synaptic transmission, through vesicle recycling staining (AM 1-43), and the numbers of ribbons (Ribeye) and synaptic vesicles (Rab 3A, RIM 1) will be provided for 2, 6 and 8 hours of hyperstimulation. The nitric oxide pathway and its relation to immediate early gene expression will also be investigated in utricular hair cells during these time-exposures to hypergravity. Investigations of these same proteins and vesicle recycling in utricular hair cells of tilted mice will determine their functional capabilities. Thus, this project will help us to understand the early and long term effects of altered gravity on the function of its primary receptors, the utricular hair cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This basic research in normal gravity and in a microgravity model of vestibular sensory deprivation gave new insights on the development and adult physiology of hair cells. Studies of Na+ and K+ voltage-dependent channels determined their precise patterns of expression in vestibular sensory epithelium from birth to adult with direct relevance to the normal functioning of hair cells (Wooltorton et al., 2006; Hurley et al., 2006; Gaboyard et al., in preparation). Lipofuscin-like organelles were investigated and quantified in hair cells of rodents in normal environmental conditions giving new insights on their potential link to aging and ototoxicity (Gaboyard and Lysakowski, in preparation). Morphological and molecular properties of macular hair cells were determined in a rodent model of sensory deprivation that corresponds to a micro-gravitational environment. This study showed that hair cells developed and living in a micro-gravitational environment display normal properties to function properly in normal gravity; this suggests a capacity of the vestibule to develop and survive in different gravitational environments (Gaboyard, in preparation).
Task Progress:	Unlike the first year, in which we mainly set up “tools” for this study, this second and last year was focused on obtaining consistent and publishable results. This decision led us to investigate principally our 3rd aim: “to determine if adult hair cells are capable of function after development in sensory deprivation”, while the 1st and 2nd aims were left unfinished. On the one hand, we realized that injection of AM 1-43 did not highlight a recycling pathway of exo/endocytosis, as we had misinterpreted (progress report 2005). The “labeled multivesicular bodies” were, in fact, lipofuscin-like organelles. Even if the identification and quantification of these autofluorescent organelles in normal gravity conditions gave new insights on rodent vestibular hair cells (Gaboyard, in preparation), these lipofuscin-like organelles were no longer relevant to our 1st aim: “to study the vesicle recycling rate of hair cells exposed to hypergravity for different time-exposures”. In hypergravitational conditions (unpublished), we observed some discrepancies between experiments in the number of lipofuscin-like organelles in utricular hair cells, however we did not have enough time left to perform standardized experiments linking these differences to physiological state (age, weight…) or gravity condition. On the other hand, unlike the first year when we could not breed our animal model of vestibular sensory deprivation, the tilted mice, this year we had enough animals to precisely investigate their macular hair cells using morphological and molecular “tools” developed during the first year (Hurley et al., 2006; Wooltorton et al., 2006; Gaboyard et al., in preparation). We found that from their hair bundles, the mechanical sensors of head movements, to their afferents, transmitters of the sensory information, macular hair cells of the tilted mice display the morphological organization and molecular composition (actin filaments, calcium-binding protein, voltage-dependent channels and synaptic proteins) observed in control mice with a normal vestibular stimulation (Gaboyard, in preparation). In summary, as in every new project, more time was needed than we expected. Thus, even if very interesting results came out from this study, parts of the initial proposal remain un-investigated. Nonetheless, we answered the question of our 3rd aim by showing that adult vestibular hair cells developing in a sensory deprived environment have the morphological and molecular capacities to function properly.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Articles in Peer-reviewed Journals	Hurley KM, Gaboyard S, Zhong M, Price SD, Wooltorton JR, Lysakowski A, Eatock RA. "M-like K+ currents in type I hair cells and calyx afferent endings of the developing rat utricle." J Neurosci. 2006 Oct 4;26(40):10253-69. PMID: 17021181 , Oct-2006
Articles in Peer-reviewed Journals	Wooltorton JR, Gaboyard S, Hurley KM, Price SD, Garcia JL, Zhong M, Lysakowski A, Eatock RA. "Developmental changes in two voltage-dependent sodium currents in utricular hair cells." J Neurophysiol. 2007 Feb;97(2):1684-704. PMID: 17065252 , Feb-2007